

- 4) Calculate the test statistic under the standard normal distribution:

$$z = \frac{21819 - 22000}{\frac{1295}{\sqrt{100}}} = -1.40$$

For 5% level of significance $-1.40 > 1.96$. Hence, we cannot reject the Null hypothesis.

Check Your Progress 2

- 1) 95% is wider.
- 2) You are 95% confident that the interval contains the parameter.
- 3) You use t when the standard error is estimated and z when the standard error is known. One exception is for a confidence interval for a proportion when z is used even though the standard error is estimated.
- 4) See Section 31.2
- 5) See Section 31.3
- 6) See Section 31.3
- 7) See Sections 31.3 and 31.4
- 8) The determination made by the analyst may be said to be unbiased if the mean value in the population, that could be obtained if she took an infinite number of readings, can be supposed to be 165^0 . We have, therefore, to test the null hypothesis $H_0: \mu = 165$ against the alternative $H_1: \mu \neq 165$.

It will be assumed (a) that the population distribution is of the normal type and (b) that the sample observations are random and mutually independent.

Under these assumptions, a test H_0 is provided by the statistic $t = \sqrt{n}(\bar{x} - 165)/s'$, which has $df = (n-1)$.

For the given observations, $n = 12$, $\bar{x} = 163.992$,

$$s' = \frac{1}{n-1} \sum_i (x_i - \bar{x})^2 = 3.039; \text{ so that } t = -1.149.$$

From t table $t_{0.025, 11} = 2.201$ and $t_{0.005, 11} = 3.106$. Since for the given sample $|t|$ is smaller than both these tabulated values, H_0 is to be accepted at both the 1% and the 5% level of significance. In other words, we find no reason to suppose that the analyst's determination is not free from bias.

Check Your Progress 3

- 1) Read standardization of normal variate and answer
- 2) $p=0.101$
- 3) see Section 20.6

- 4) Here we have to test $H_0: \mu_1 = \mu_2$ against the alternative $H: \mu_1 \neq \mu_2$. The test for H_0 is then provided by the statistic $t = \frac{(\bar{x}_1 - \bar{x}_2)}{s' \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$ which is t with $df = n_1 + n_2 - 2$, $t_{0.025, 13} = 2.160$, $t_{0.005, 13} = 3.012$.

Check Your Progress 4

- 1) The null hypothesis here is $H_0: \rho = 0$, to be tested against all alternatives. As we have seen, under certain assumptions, which may be considered legitimate here, the test is given by $t = r \sqrt{n-2} / \sqrt{1-r^2}$ which has $df = n - 2$. Here $t = 0.880$ and tabulated values are $t_{0.025, 18} = 2.101$ and $t_{0.005, 18} = 2.878$. The observed value is, therefore, insignificant at both the levels; i.e. the population correlation coefficient may be supposed to be zero.
- 2) (a) 50%; (b) 8.08%; (c) 44.35
- 3) (a) 47.25; (b) 34.59
- 4) 83.46 and 96.4

31.11 EXERCISES

Q1. The weights at birth for 15 babies born are given below. Each figure is correct to the nearest tenth of a pound.

6.2, 5.7, 8.1, 6.7, 4.8, 5.0, 7.1, 6.8, 5.8, 6.9, 7.6, 7.9, 7.5, 7.8, 8.5.

Give two limits between which the mean weight at birth for all such babies is likely to lie.

Ans. Let us denote by x the variable: weight at birth per baby. Our problem here is then to find, on the basis of the given sample of 15 babies, confidence limits for the population mean of x . We shall assume (a) that in the population x is normally distributed (with a mean μ and standard deviation σ , both of which are unknown) and (b) that the given observations are a random sample from the distribution.

Under these assumptions, the $100(1-\alpha) \%$ confidence limits to μ will be

$$(\bar{x} - t_{\alpha/2, n-1}) \frac{s'}{\sqrt{n}} \text{ and } (\bar{x} + t_{\alpha/2, n-1}) \frac{s'}{\sqrt{n}}$$

Q2. It has been said by some educationists that mathematical ability varies widely. To examine this suggestion, 15 students of class IX are given a mathematical aptitude test carrying 100 marks. The score of the students on the test are shown below:

73, 16, 84, 20, 13, 63, 53, 68, 19, 40, 52, 91, 12, 25, 17.

Examine the more specific suggestion that the standard deviation of score per student is higher than 20.

Ans. We shall assume that the random variable x , viz. score per student on the test is distributed normally for students of class IX with some mean μ and standard deviation σ , both unknown. Further the given set of observations will be regarded as the observed values for a random sample of size $n=15$ from this distribution. So, the problem can be stated as $H_0: \sigma = 20$ against the alternative $H: \sigma > 20$. Under the usual assumption, the test is given by

$$\chi^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{\sigma_0^2}. \text{ The value of } \chi_{0.05, 14}^2 = 23.685.$$

Q3. Two experimenters, A and B, take repeated measurements on the length of a copper wire. On the basis of the data obtained by them, which are given below, test whether B's measurements are more accurate than A's. (It may be supposed that the readings taken by both are unbiased.)

A's measurements: 12.47, 12.44, 11.90, 12.13, 12.77, 11.86, 11.96, 12.25, 12.78, 12.29.

B's measurements: 12.06, 12.34, 12.23, 12.46, 12.39, 12.46, 11.98, 12.22.

Ans. The problem can be stated as $H_0: \sigma_1 = \sigma_2$, against the alternative $H: \sigma_1 > \sigma_2$.

Under the usual assumption, the test is given by $F = \frac{s_1'^2}{s_2'^2}$ with $(n_1 - 1, n_2 - 1)$

degrees of freedom. The tabulated values are $F_{0.05; 9, 7} = 3.68$; $F_{0.01; 9, 7} = 6.72$.

Q4. In a certain industrial experiment, a job was performed by 15 workmen according to a particular method (say, Method I) and by 15 other workmen according to a second method (Method II). The time (in minutes) taken by each workman to complete the job is shown below:

Method I: 55, 53, 57, 55, 52, 51, 54, 54, 53, 56, 50, 54, 52, 56, 51.

Method II: 54, 53, 56, 60, 58, 55, 56, 54, 58, 57, 55, 54, 59, 52, 54.

Test if there was difference of time taken between these two methods.

Ans. The problem can be stated as $H_0: \mu_1 = \mu_2$, against the alternative $H: \mu_1 < \mu_2$.

Under the usual assumption, the test is given by $t = \frac{(\bar{x}_1 - \bar{x}_2)}{s' \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$ with $(n_1 + n_2$

$- 2)$ degrees of freedom.

The tabulated values are $-t_{0.05, 28} = -1.701$; $-t_{0.01, 28} = -2.467$.]

Q5. The marks in mathematics (x) and those in English (y) are given below for a group of 14 students:

x : 52, 31, 83, 75, 95, 78, 85, 23, 69, 32, 48, 9, 84, 54.

y: 69, 42, 50, 31, 43, 38, 59, 44, 51, 61, 33, 43, 27, 46.

These will be used to examine the claim of some educationists that mathematical ability and proficiency in English are inversely related (i.e. negatively correlated).

Ans. The problem can be stated as $H_0: \rho=0$, against the alternative $H: \rho < 0$.

Under certain assumptions, the test is given by $t = r \sqrt{n-2} / \sqrt{1-r^2}$ which has $df = n-2$. Here $t = -0.495$ and the tabulated value is $t_{0.05, 12} = -1.782$, H_0 is to be accepted at the 5% level of significance. In other words, we find no evidence in the data to support the claim that x and y are negatively correlated.

Q6. The weights (in lb.) of 10 boys before they are subjected to a change of diet and after a lapse of six months are recorded below:

Before: 109, 112, 98, 114, 102, 97, 88, 101, 89, and 91.

After: 115, 120, 99, 117, 105, 98, 91, 99, 93, and 89.

Test whether there is any significant gain in weight as a result of the change in diet.

Ans. The problem can be stated as $H_0: \mu_x = \mu_y$, against the alternative $H: \mu_x > \mu_y$.

Under certain assumptions, the test is given by $t = \sqrt{n} \bar{z} / s'_z$ with $df = n - 1$, where $z = x - y$.

Here, $\bar{z} = 2.5$, $s'_z = 3.171$, $t = 2.493$ and the tabulated value is $t_{0.05, 9} = 1.833$ and $t_{0.01, 9} = 2.821$. The observed value is thus significant at the 5% but insignificant at the 1% level of significance. If we choose the 5% level, then the null hypothesis should be rejected and we should say that the change of diet results in a gain in average weight.

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STATISTICAL TABLES

Table A1: Normal Area Table

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974

2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990

Table A2: Critical Values of Chi-squared Distribution

df\area	0.1	0.05	0.025	0.01	0.005
1	2.706	3.841	5.024	6.635	7.879
2	4.605	5.991	7.378	9.210	10.597
3	6.251	7.815	9.348	11.345	12.838
4	7.779	9.488	11.143	13.277	14.860
5	9.236	11.071	12.833	15.086	16.750
6	10.645	12.592	14.449	16.812	18.548
7	12.017	14.067	16.013	18.475	20.278
8	13.362	15.507	17.535	20.090	21.955
9	14.684	16.919	19.023	21.666	23.589
10	15.987	18.307	20.483	23.209	25.188
11	17.275	19.675	21.920	24.725	26.757
12	18.549	21.026	23.337	26.217	28.300
13	19.812	22.362	24.736	27.688	29.819
14	21.064	23.685	26.119	29.141	31.319
15	22.307	24.996	27.488	30.578	32.801
16	23.542	26.296	28.845	32.000	34.267
17	24.769	27.587	30.191	33.409	35.718
18	25.989	28.869	31.526	34.805	37.156
19	27.204	30.144	32.852	36.191	38.582
20	28.412	31.410	34.170	37.566	39.997
21	29.615	32.671	35.479	38.932	41.401
22	30.813	33.924	36.781	40.289	42.796
23	32.007	35.172	38.076	41.638	44.181
24	33.196	36.415	39.364	42.980	45.559
25	34.382	37.652	40.646	44.314	46.928
26	35.563	38.885	41.923	45.642	48.290
27	36.741	40.113	43.195	46.963	49.645
28	37.916	41.337	44.461	48.278	50.993
29	39.087	42.557	45.722	49.588	52.336
30	40.256	43.773	46.979	50.892	53.672

Table A3: Critical Values of t Distribution

Df/p	0.25	0.10	0.05	0.025	0.01	0.005
1	1.0000	3.0777	6.3138	12.7062	31.8205	63.6567
2	0.8165	1.8856	2.9200	4.3027	6.9646	9.9248
3	0.7649	1.6377	2.3534	3.1825	4.5407	5.8409
4	0.7407	1.5332	2.1318	2.7765	3.7470	4.6041
5	0.7267	1.4759	2.0150	2.5706	3.3649	4.0321
6	0.7176	1.4398	1.9432	2.4469	3.1427	3.7074
7	0.7111	1.4149	1.8946	2.3646	2.9980	3.4995
8	0.7064	1.3968	1.8595	2.3060	2.8965	3.3554
9	0.7027	1.3830	1.8331	2.2622	2.8214	3.2498
10	0.6998	1.3722	1.8125	2.2281	2.7638	3.1693
11	0.6974	1.3634	1.7959	2.2010	2.7181	3.1058
12	0.6955	1.3562	1.7823	2.1788	2.6810	3.0545
13	0.6938	1.3502	1.7709	2.1604	2.6503	3.0123
14	0.6924	1.3450	1.7613	2.1448	2.6245	2.9768
15	0.6912	1.3406	1.7531	2.1315	2.6025	2.9467
16	0.6901	1.3368	1.7459	2.1199	2.5835	2.9208
17	0.6892	1.3334	1.7396	2.1098	2.5669	2.8982
18	0.6884	1.3304	1.7341	2.1009	2.5524	2.8784
19	0.6876	1.3277	1.7291	2.0930	2.5395	2.8609
20	0.6870	1.3253	1.7247	2.0860	2.5280	2.8453
20	0.6870	1.3253	1.7247	2.0860	2.5280	2.8453
21	0.6864	1.3232	1.7207	2.0796	2.5177	2.8314
22	0.6858	1.3212	1.7171	2.0739	2.5083	2.8188
23	0.6853	1.3195	1.7139	2.0687	2.4999	2.8073
24	0.6849	1.3178	1.7109	2.0639	2.4922	2.7969
25	0.6844	1.3163	1.7081	2.0595	2.4851	2.7874
26	0.6840	1.3150	1.7056	2.0555	2.4786	2.7787
27	0.6837	1.3137	1.7033	2.0518	2.4727	2.7707
28	0.6834	1.3125	1.7011	2.0484	2.4671	2.7633
29	0.6830	1.3114	1.6991	2.0452	2.4620	2.7564
30	0.6828	1.3104	1.6973	2.0423	2.4573	2.7500
inf	0.6745	1.2816	1.6449	1.9600	2.3264	2.5758

Table A4: Critical Values of *F* Distribution

(5% level of significance)

df2/df1	1	2	3	4	5	6	7	8	9	10
1	161.448	199.500	215.707	224.583	230.162	233.986	236.768	238.883	240.543	241.882
2	18.513	19.000	19.164	19.247	19.296	19.330	19.353	19.371	19.385	19.396
3	10.128	9.552	9.277	9.117	9.014	8.941	8.887	8.845	8.812	8.786
4	7.709	6.944	6.591	6.388	6.256	6.163	6.094	6.041	5.999	5.964
5	6.608	5.786	5.410	5.192	5.050	4.950	4.876	4.818	4.773	4.735
6	5.987	5.143	4.757	4.534	4.387	4.284	4.207	4.147	4.099	4.060
7	5.591	4.737	4.347	4.120	3.972	3.866	3.787	3.726	3.677	3.637
8	5.318	4.459	4.066	3.838	3.688	3.581	3.501	3.438	3.388	3.347
9	5.117	4.257	3.863	3.633	3.482	3.374	3.293	3.230	3.179	3.137
10	4.965	4.103	3.708	3.478	3.326	3.217	3.136	3.072	3.020	2.978
11	4.844	3.982	3.587	3.357	3.204	3.095	3.012	2.948	2.896	2.854
12	4.747	3.885	3.490	3.259	3.106	2.996	2.913	2.849	2.796	2.753
13	4.667	3.806	3.411	3.179	3.025	2.915	2.832	2.767	2.714	2.671
14	4.600	3.739	3.344	3.112	2.958	2.848	2.764	2.699	2.646	2.602
15	4.543	3.682	3.287	3.056	2.901	2.791	2.707	2.641	2.588	2.544
16	4.494	3.634	3.239	3.007	2.852	2.741	2.657	2.591	2.538	2.494
17	4.451	3.592	3.197	2.965	2.810	2.699	2.614	2.548	2.494	2.450
18	4.414	3.555	3.160	2.928	2.773	2.661	2.577	2.510	2.456	2.412
19	4.381	3.522	3.127	2.895	2.740	2.628	2.544	2.477	2.423	2.378
20	4.351	3.493	3.098	2.866	2.711	2.599	2.514	2.447	2.393	2.348
21	4.325	3.467	3.073	2.840	2.685	2.573	2.488	2.421	2.366	2.321
22	4.301	3.443	3.049	2.817	2.661	2.549	2.464	2.397	2.342	2.297
23	4.279	3.422	3.028	2.796	2.640	2.528	2.442	2.375	2.320	2.275
24	4.260	3.403	3.009	2.776	2.621	2.508	2.423	2.355	2.300	2.255
25	4.242	3.385	2.991	2.759	2.603	2.490	2.405	2.337	2.282	2.237
26	4.225	3.369	2.975	2.743	2.587	2.474	2.388	2.321	2.266	2.220
27	4.210	3.354	2.960	2.728	2.572	2.459	2.373	2.305	2.250	2.204
28	4.196	3.340	2.947	2.714	2.558	2.445	2.359	2.291	2.236	2.190
29	4.183	3.328	2.934	2.701	2.545	2.432	2.346	2.278	2.223	2.177
30	4.171	3.316	2.922	2.690	2.534	2.421	2.334	2.266	2.211	2.165
40	4.085	3.232	2.839	2.606	2.450	2.336	2.249	2.180	2.124	2.077
60	4.001	3.150	2.758	2.525	2.368	2.254	2.167	2.097	2.040	1.993
120	3.920	3.072	2.680	2.447	2.290	2.175	2.087	2.016	1.959	1.911
inf	3.842	2.996	2.605	2.372	2.214	2.099	2.010	1.938	1.880	1.831

Table A4: Critical Values of F Distribution (Contd.)

(5% level of significance)

df2/df1	12	15	20	24	30	40	60	120	INF
1	243.906	245.950	248.013	249.052	250.095	251.143	252.196	253.253	254.314
2	19.413	19.429	19.446	19.454	19.462	19.471	19.479	19.487	19.496
3	8.745	8.703	8.660	8.639	8.617	8.594	8.572	8.549	8.526
4	5.912	5.858	5.803	5.774	5.746	5.717	5.688	5.658	5.628
5	4.678	4.619	4.558	4.527	4.496	4.464	4.431	4.399	4.365
6	4.000	3.938	3.874	3.842	3.808	3.774	3.740	3.705	3.669
7	3.575	3.511	3.445	3.411	3.376	3.340	3.304	3.267	3.230
8	3.284	3.218	3.150	3.115	3.079	3.043	3.005	2.967	2.928
9	3.073	3.006	2.937	2.901	2.864	2.826	2.787	2.748	2.707
10	2.913	2.845	2.774	2.737	2.700	2.661	2.621	2.580	2.538
11	2.788	2.719	2.646	2.609	2.571	2.531	2.490	2.448	2.405
12	2.687	2.617	2.544	2.506	2.466	2.426	2.384	2.341	2.296
13	2.604	2.533	2.459	2.420	2.380	2.339	2.297	2.252	2.206
14	2.534	2.463	2.388	2.349	2.308	2.266	2.223	2.178	2.131
15	2.475	2.403	2.328	2.288	2.247	2.204	2.160	2.114	2.066
16	2.425	2.352	2.276	2.235	2.194	2.151	2.106	2.059	2.010
17	2.381	2.308	2.230	2.190	2.148	2.104	2.058	2.011	1.960
18	2.342	2.269	2.191	2.150	2.107	2.063	2.017	1.968	1.917
19	2.308	2.234	2.156	2.114	2.071	2.026	1.980	1.930	1.878
20	2.278	2.203	2.124	2.083	2.039	1.994	1.946	1.896	1.843
21	2.250	2.176	2.096	2.054	2.010	1.965	1.917	1.866	1.812
22	2.226	2.151	2.071	2.028	1.984	1.938	1.889	1.838	1.783
23	2.204	2.128	2.048	2.005	1.961	1.914	1.865	1.813	1.757
24	2.183	2.108	2.027	1.984	1.939	1.892	1.842	1.790	1.733
25	2.165	2.089	2.008	1.964	1.919	1.872	1.822	1.768	1.711
26	2.148	2.072	1.990	1.946	1.901	1.853	1.803	1.749	1.691
27	2.132	2.056	1.974	1.930	1.884	1.836	1.785	1.731	1.672
28	2.118	2.041	1.959	1.915	1.869	1.820	1.769	1.714	1.654
29	2.105	2.028	1.945	1.901	1.854	1.806	1.754	1.698	1.638
30	2.092	2.015	1.932	1.887	1.841	1.792	1.740	1.684	1.622
40	2.004	1.925	1.839	1.793	1.744	1.693	1.637	1.577	1.509
60	1.917	1.836	1.748	1.700	1.649	1.594	1.534	1.467	1.389
120	1.834	1.751	1.659	1.608	1.554	1.495	1.429	1.352	1.254
inf	1.752	1.666	1.571	1.517	1.459	1.394	1.318	1.221	1.000